

Units of measurement

A unit of measurement is a quantity used as a standard, allowing values of things to be compared. These include seconds (time), meters (length), and kilograms (mass). Two widely used systems of measurement are the metric system and the imperial system.

AREA	
metric	
100 square millimeters (mm ²)	= 1 square centimeter (cm ²)
10,000 square centimeters (cm ²)	= 1 square meter (m ²)
10,000 square meters (m ²)	= 1 hectare (ha)
100 hectares (ha)	= 1 square kilometer (km ²)
1 square kilometer (km ²)	= 1,000,000 square meters (m ²)
imperial	
144 square inches (sq in)	= 1 square foot (sq ft)
9 square feet (sq ft)	= 1 square yard (sq yd)
1,296 square inches (sq in)	= 1 square yard (sq yd)
43,560 square feet (sq ft)	= 1 acre
640 acres	= 1 square mile (sq mile)

LIQUID VOLUME	
metric	
1,000 milliliters (ml)	= 1 liter (l)
100 liters (l)	= 1 hectoliter (hl)
10 hectoliters (hl)	= 1 kiloliter (kl)
1,000 liters (l)	= 1 kiloliter (kl)
imperial	
8 fluid ounces (fl oz)	= 1 cup
20 fluid ounces (fl oz)	= 1 pint (pt)
4 gills (gi)	= 1 pint (pt)
2 pints (pt)	= 1 quart (qt)
4 quarts (qt)	= 1 gallon (gal)
8 pints (pt)	= 1 gallon (gal)

MASS	
metric	
1,000 milligrams (mg)	= 1 gram (g)
1,000 grams (g)	= 1 kilogram (kg)
1,000 kilograms (kg)	= 1 tonne (t)
imperial	
16 ounces (oz)	= 1 pound (lb)
14 pounds (lb)	= 1 stone
112 pounds (lb)	= 1 hundredweight
20 hundredweight	= 1 ton

LENGTH	
metric	
10 millimeters (mm)	= 1 centimeter (cm)
100 centimeters (cm)	= 1 meter (m)
1,000 millimeters (mm)	= 1 meter (m)
1,000 meters (m)	= 1 kilometer (km)
imperial	
12 inches (in)	= 1 foot (ft)
3 feet (ft)	= 1 yard (yd)
1,760 yards (yd)	= 1 mile
5,280 feet (ft)	= 1 mile
8 furlongs	= 1 mile

TIME	
metric and imperial	
60 seconds	= 1 minute
60 minutes	= 1 hour
24 hours	= 1 day
7 days	= 1 week
52 weeks	= 1 year
1 year	= 12 months

TEMPERATURE			
	Fahrenheit	Celsius	Kelvin
Boiling point of water	= 212°	100°	373°
Freezing point of water	= 32°	0°	273°
Absolute zero	= -459°	-273°	0°